

RescueVision

AI-Powered Hazard Zone Person Detection

Technical Documentation · backprop team · May 2026

Repository

<https://github.com/fradane/backprop-rescueVision>

RescueVision is a real-time edge-AI system that uses the Luxonis OAK-4 depth camera and the DepthAI SDK to detect people in hazardous areas, geo-locate them via stereo depth, monitor their motion status, and classify their body posture — providing actionable alerts when someone may be injured or incapacitated.

1. Introduction

Hazardous environments — industrial sites, disaster zones, remote terrain — pose serious risks to human life. Early detection of distressed or immobile persons can be the difference between a successful rescue and a fatality. RescueVision addresses this gap by deploying an intelligent, self-contained vision node that continuously monitors a scene and raises precise, geo-referenced alerts.

The system is built around four core capabilities:

- Person detection via a neural inference pipeline running on the OAK-4 onboard processor.
- Depth-assisted geo-location that converts pixel coordinates + stereo depth into real-world latitude/longitude.
- Immobility detection that flags individuals who have remained stationary beyond a configurable time threshold.
- Posture classification that distinguishes upright (standing/walking) persons from prone (lying down) persons.

Together, these capabilities enable first responders and safety operators to receive structured, map-ready alerts with minimal human monitoring effort.

2. Hardware — Luxonis OAK-4 Camera

The OAK-4 (OpenCV AI Kit, 4th generation) is a production-grade edge-AI camera manufactured by Luxonis. It integrates an RGB sensor, stereo depth cameras, and an Intel Myriad X (or equivalent) Vision Processing Unit (VPU) into a single enclosure, enabling complex inference pipelines to run entirely on-device without a GPU server.

Manufacturer	Luxonis
Model	OAK-4
Stereo baseline	~7.5 cm (calibrated)
Depth range	0.2 m – 35 m (active stereo)
RGB resolution	Up to 4K @ 30 fps
On-device AI	Intel Myriad X VPU / RVC4 NPU
SDK	DepthAI Python SDK
Interface	USB 3.1 / PoE (model dependent)

Running the entire inference pipeline on the OAK-4 VPU minimises USB bandwidth requirements and reduces host-CPU load, making the system viable on lightweight edge computers (e.g. Raspberry Pi 5, NVIDIA Jetson Orin Nano).

3. System Architecture

RescueVision follows a pipelined, single-pass architecture. Each frame from the OAK-4 passes through a sequence of stages, each enriching the data before it is consumed by the next.

Stage 1 — Person Detection	YOLOv8 / MobileNet-SSD compiled for Myriad X detects all persons in the frame. Bounding boxes and confidence scores are returned.
Stage 2 — Depth Extraction	DepthAI SpatialDetectionNetwork fuses RGB detections with the stereo depth map to produce XYZ coordinates (in metres) for each detected person.
Stage 3 — Geo-location	Camera GPS fix (or known static mount position) is combined with the XYZ offset to compute absolute latitude/longitude for each detection.
Stage 4 — Track & Immobility	A lightweight centroid tracker maintains per-person tracks. A timer is incremented for persons whose spatial position does not change beyond a configurable threshold. Alerts fire when the threshold is exceeded.
Stage 5 — Posture Classification	The bounding-box aspect ratio and an optional skeleton keypoint model are used to classify each person as STANDING or PRONE. Prone persons trigger a priority alert.
Stage 6 — Alert Dispatch	Structured JSON events (person ID, GPS coordinates, posture, immobility flag) are published over a local REST/WebSocket endpoint for consumption by a dashboard or external system.

4. Person Detection

4.1 Model

A person-detection model (YOLOv8n or MobileNet-SSD-based) is compiled to the OpenVINO IR format and loaded onto the OAK-4 VPU via the DepthAI MobileNet/YOLO detection node. Only detections whose label corresponds to the 'person' class and whose confidence exceeds a configurable threshold (default: 0.5) are forwarded downstream.

4.2 DepthAI Pipeline

The DepthAI pipeline is constructed in Python using the depthai library. Key nodes:

- ColorCamera — streams RGB frames at 1080p/30 fps.

- StereoDepth — fuses left/right mono cameras into a dense depth map.
- SpatialDetectionNetwork — runs detection inference and enriches each bounding box with the XYZ centroid from the depth map.
- XLinkOut — streams results to the host for tracking, classification, and alerting.

5. Depth Estimation and Geo-Location

5.1 Stereo Depth

DepthAI's onboard stereo engine produces a calibrated depth map aligned with the RGB frame. The SpatialDetectionNetwork reports, for each bounding box, the spatial coordinates (X, Y, Z) of the detection centroid in the camera reference frame (metres).

5.2 Coordinate Conversion

The OAK-4 outputs, for each detected person, their position in 3D space relative to the camera — essentially telling us how many metres left/right, up/down, and in front of the lens the person is standing. To turn that relative position into a usable GPS coordinate, the system needs two pieces of additional information: where the camera itself is placed in the world (its own latitude and longitude), and which direction it is pointing (its compass heading).

With those two inputs the conversion is straightforward: the metric offset between the camera and the detected person is translated into a small shift in latitude and longitude. The result is an absolute GPS coordinate for each person in the scene, ready to be dropped onto any map or forwarded to a dispatch system.

In practice

If the camera is mounted on a fixed structure (e.g. a mast or tripod), its GPS position and heading are configured once at deployment time. If it is carried by a drone or a responder, a GPS module and compass supply the values in real time.

6. Immobility Detection

6.1 Centroid Tracking

Each detected person is assigned a unique track ID via a simple Euclidean centroid tracker. Between consecutive frames the tracker associates new detections to existing tracks based on minimum centroid distance. New detections that cannot be matched (distance > max_distance threshold) start a new track.

6.2 Stillness Logic

For each active track, the 3D spatial position (X, Y, Z) is compared against the position recorded at the start of the current stillness window. A person is classified as IMMOBILE when:

- The Euclidean distance between the current and last-known 3D position is below the movement threshold (default: 0.15 m).
- This condition has been met continuously for longer than the immobility timeout (default: 10 s, configurable).

Configurable Parameters

movement_threshold (metres, default 0.15) | immobility_timeout (seconds, default 10) | max_distance (tracker association, default 1.0 m)

7. Posture Classification

7.1 Bounding-Box Aspect Ratio

For each detected person the width and height of the bounding box are compared. A standing or walking person occupies a tall, narrow box (height > width), while a person lying down produces a wide, flat box (width > height). When the ratio flips — width exceeds height — the system classifies the person as PRONE and raises a priority alert.

STANDING	Bounding box height > width. Person is upright.
PRONE	Bounding box width > height. Person is lying down. Triggers PRIORITY ALERT.

8. Alert System

Alerts are structured JSON payloads published to a local REST endpoint (POST /alerts) and optionally forwarded to a remote dashboard. Each alert contains:

track_id	Unique integer identifying the person track
timestamp	ISO-8601 UTC timestamp
latitude	Decimal degrees, WGS-84
longitude	Decimal degrees, WGS-84
posture	STANDING PRONE
immobile	true false
immobile_seconds	Duration in seconds (0 if mobile)
confidence	Detection confidence score (0.0 – 1.0)
alert_level	INFO WARNING CRITICAL

Alert level logic:

- INFO — person detected, mobile, standing.
- WARNING — person immobile beyond timeout OR posture is PRONE.
- CRITICAL — person is both immobile beyond timeout AND posture is PRONE.

9. Software Dependencies

depthai	DepthAI Python SDK — OAK-4 pipeline management and inference
opencv-python	Frame visualisation and bounding-box rendering
numpy	Numerical operations, coordinate transforms
scipy	Optional: Hungarian algorithm for robust tracking
fastapi / uvicorn	REST endpoint for alert dispatch
pydantic	Alert schema validation
pyserial	GPS NMEA sentence parsing (if hardware GPS attached)

10. Setup and Usage

10.1 Prerequisites

- Luxonis OAK-4 camera connected via USB 3.1.
- Python 3.9+ environment.
- DepthAI SDK installed: `pip install depthai`.
- (Optional) GPS module connected via serial for dynamic geo-location.

10.2 Installation

```
git clone https://github.com/fradane/backprop-rescueVision.git
cd backprop-rescueVision
pip install -r requirements.txt
```

10.3 Running

```
python main.py --lat 45.186 --lon 7.875 --heading 0 --timeout 10
```

Key CLI flags:

- `--lat / --lon` — camera mount coordinates (decimal degrees).
- `--heading` — camera bearing in degrees from north (0 = north-facing).

- `--timeout` — immobility alert threshold in seconds (default: 10).
- `--threshold` — minimum movement in metres to reset immobility timer (default: 0.15).
- `--port` — REST server port (default: 8000).

11. Conclusions

RescueVision demonstrates that a single OAK-4 camera, combined with the DepthAI SDK and a lightweight software stack, can deliver a complete, geo-aware, real-time person safety monitor capable of running on commodity edge hardware. The four-stage pipeline — detection, geo-location, immobility tracking, and posture classification — provides the structured situational awareness that emergency responders need, without relying on external cloud services or heavy GPU infrastructure.

The project is open-source and designed to be extended. We welcome contributions from the community to improve model accuracy, expand hardware support, and deepen integration with existing emergency response platforms.

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